

A Survey on IOT Based Smart Helmet for Construction Workers

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Abstract

The construction industry has always been the most arduous area to work in because of risks associated with the health of the workers. Government has made it mandatory for workers to wear helmets when working on sites as ensuring safety of workers is important. But the regular helmet does not make big change when it comes to health-related hazards caused while working on sites. This review paper aims to provide an overview of the advancements in Helmet technologies, along with their functionalities and effectiveness. This smart wearable works on immaculate connectivity accredited by IOT protocols, where real time data is transmitted for analysis, this real time data transfer will enhance communication. Various sensors are used which will help in continuous monitoring health and safety of workers. This system will ensure safety and create awareness between the people working on the site and avoid accidents which will prevent number of deaths on construction sites. This review highlights the development of safety helmets that will revolutionize workplace safety within and beyond the construction industry. Through a comprehensive analysis of its potential, limitations and future prospects, this article aims to inform researchers and policy makers about the role of smart technology in creating safe work place and reducing hazard.

Keywords: Smart helmet, IOT protocols, Sensors, Microcontroller, communication technology, VLSI

INTRODUCTION

The construction site has a huge workforce, and most of the work is done manually. But some of the work comes along with a risk of life as workers are exposed to various hazards such as falls, collisions, and also exposure to harmful substances. As per the law issued by government it is important to ensure that all the people working on a construction site, wear safety helmets and shoes, as without taking preventive measures, the probability of accidents increases. To minimize such unpredictable accidents, safety measures need to be taken hence various systems are implemented for same. With the advent of

IOT technology there has been a fundamental change in safety management practices, offering real-time monitoring, data analytics, and predictive capabilities, leading to the development of smart solutions like IoT- based smart helmets.

These helmets provide a dynamic approach towards accident prevention. In this survey paper, we aim to provide an overview of IoT-based smart helmets for people working at construction sites, exploring their functionalities, features, applications, challenges Through scrutiny of relevant literature, case studies, and real-world implementations, we seek to highlight the transformative impact of smart helmet technology on construction site safety, productivity and identify opportunities for further research and innovation in this field.

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LITERATURE SURVEY

“Hard Hat Detection for Construction Safety Visualization” Kishor Shrestha, et. They used image processing techniques to check whether the worker is wearing a helmet or not. But it does not provide any safety measures to overcome in any emergency conditions [1].

“Smart Helmet for Coal Miners” V. Sai Prasanna Kumar, et. This system ensures the safety of workers while they are working in the mines as it uses LoRa communication technology and GSM to transmit and receive data. It is composed of various sensors for analysis of certain parameters. MQ2 sensor is used to identify harmful gases like SO₂, CO₂, NO₂, NH₃. Once these gases are identified data is transmitted used LoRa communication technique [2].

“Intelligent helmet for coal miners using Zigbee technique” Garima Tiwari, et. This system major motive is to provide wireless communication and monitoring systems. This system uses sensors that monitor harmful gases, humidity level in the air, abnormal temperature, etc. Data from these sensors is sent to the control room using microcontroller and this data is identified using RS232 terminal and is displayed on the monitor screen [3].

“Smart helmet and insole sensors for near fall recognition during descent of stairs” Wang C et. Construction sites consist of staircases and many times workers have to face an accident as they fall from those staircases. This system basically detects human fall by using sensors like accelerometer sensor and textile pressure sensor these sensors are attached to the helmet and the shoes of the workers, which will analyse whole body balance, head tilt pattern and planter pressure distribution. 10 physically fit men participated in the experimental survey of this system, this system have shown good performance in terms of weight-shifting patterns in the head and the feet during stair descent [4].

“Internet of things based smart helmet for construction” Preeti Kuhar et. This system modified a regular helmet into a smart one by adding a feature of work mode(on/off) to it along with a task completion button. This work mode system is used to check if a particular worker is engaged with work or not. This smart helmet is also composed of GPS, to address the exact location of any worker on site in any emergency [5].

“Analysis of the Influence factor of the safety helmet” Wu Hao et Safety helmet provide protection along with that prevent and control collision damage. But these helmets are uncomfortable so many workers avoid wearing helmets. And according to the reports 15% of the injuries take place because workers are not wearing helmets. This paper also does a detailed study on the comfortable design of the helmets and how these hard helmets can turn into comfortable helmet [6].

“Smart Helmet 5.0” Israel Campero-Jurado et. The actual reason of invention of this equipment is to improve (OHS)occupational health and safety, improve employee performance by reducing the likelihood of illness, injury or death. Another goal is to implement a smart machine at workplaces to avoid risks by detecting them early. This system consists of few platforms like hardware platform which is composed of sensors like gas, accelerometer, etc. Second platform is Intelligence module which is an open source IOT platform for data transmission, collection, processing etc. Later these platforms are evaluated using SVM, Naive Bayes Classifier, Static neural network [7]. “Sustainable helmet for coal miners” T. N. P. Madhuri et. This system aims to provide innovative monitoring system. Here microcontroller is integrated with sensors. It has buzzers for alerts in case of accidents. This buzzer provides immediate audible alerts. If accidents occur they are known by other workers and staff immediately and hence instant medical help is provided to the worker [8]. “IoT based Smart Helmet for Automated and Multi-parametric Monitoring for Underground Miners’ Health Hazards” Suwarna Karna et This paper defines development of an IOT based smart helmet for real time monitoring of worker working in mines for their health’s safety. The system is comprised of array of ambient and flexible body sensors which are wirelessly connected to cloud via LoRa WAN protocol which sets up

communication in the control room [9]. “Smart construction safety helmet” Vikas Suresh et al This system is a standard safety helmet which is integrated with sensors such as infrared sensor, temperature sensor, sound sensor, gas sensor, all these sensors are used to detect respective parameters. Arduino is connected to Bluetooth model to read the data received from sensors. Along with the safety this smart helmet is also programmed to create and manage employee payroll information [10].

This flowchart illustrates in Figure 1, the sequence of actions in the operation of a smart helmet system designed for use in construction sites. The smart helmet integrates various sensors and communication technologies to enhance safety and efficiency in construction operations.

1. *Start*: The process begins when the smart helmet is powered on.
2. *Sensor Initialization*: The system initializes and calibrates the sensors embedded in the helmet, such as proximity sensors, temperature sensors, and accelerometers.
3. *Data Acquisition*: The helmet collects data from the sensors in real-time, of environmental conditions, worker's health, etc.
4. *Data Processing*: The acquired data is processed by microcontroller or processing unit to analyze the information and identify potential hazards.
5. *Hazard Detection*: Based on the processed data, the system identifies hazards such as high temperatures, dangerous gases, or human fall detection
6. *Alert Generation*: If no hazards are detected, the system continues monitoring. If a hazard is detected then there are two types of alerts a visual alert in which helmet's LED lights flash to alert the wearer, and the second one is an auditory alert where an alarm or warning sound is activated.
7. *Communication*: The system sends an alert to nearby workers or supervisors on the site via wireless communication, along with this the helmet may also transmit the wearer's location for quick response.
8. *End*: The process ends when the helmet is powered off or the operation is completed.

ADVANTAGES

These are few major advantages of this system, this smart helmet provides real time monitoring of workers health, environmental conditions, poisonous gases, permitting immediate action in case of emergencies [2, 9]. System also detects human fall using accelerometer sensor or gyroscope sensor. And immediately alert is sent to the control room [4]. Incorporation of communication technologies like GSM, LoRa, Zigbee provides impeccable communication between workers and control room, allowing instant response to the emergencies [3, 9]. For processing the data obtained from the sensors system uses advanced data analytics technique, providing perception of workers behaviour, safety compliance, environmental conditions. This data can be utilized for dynamic safety management [7, 10].

LIMITATIONS

1. *Lack of Integration*: Many of the above systems operate on discrete solutions without integrating with wide safety management system. This lack of integration may restrict their effectiveness in providing comprehensive safety solutions.
2. *Reliability and Accuracy*: There is narrow information on the reliability and accuracy of the systems. It's important for safety systems to provide accurate and reliable data to ensure timely mediation and precaution of accidents.
3. *Comfort*: As discussed comfort is also an advantage but only in few systems, not all systems are focusing on providing a helmet with comfort, there is discomfort in maximum cases which leads to non acceptance of helmets by workers. Because of this discomfort workers avoid wearing helmet.
4. *Cost and accessibility*: Small scale construction sites or mines may not be able to provide workers with these smart helmets as they are more in cost as compared to the usual helmet. They might also not have accessibility in case of any technical issue occurred in helmet.

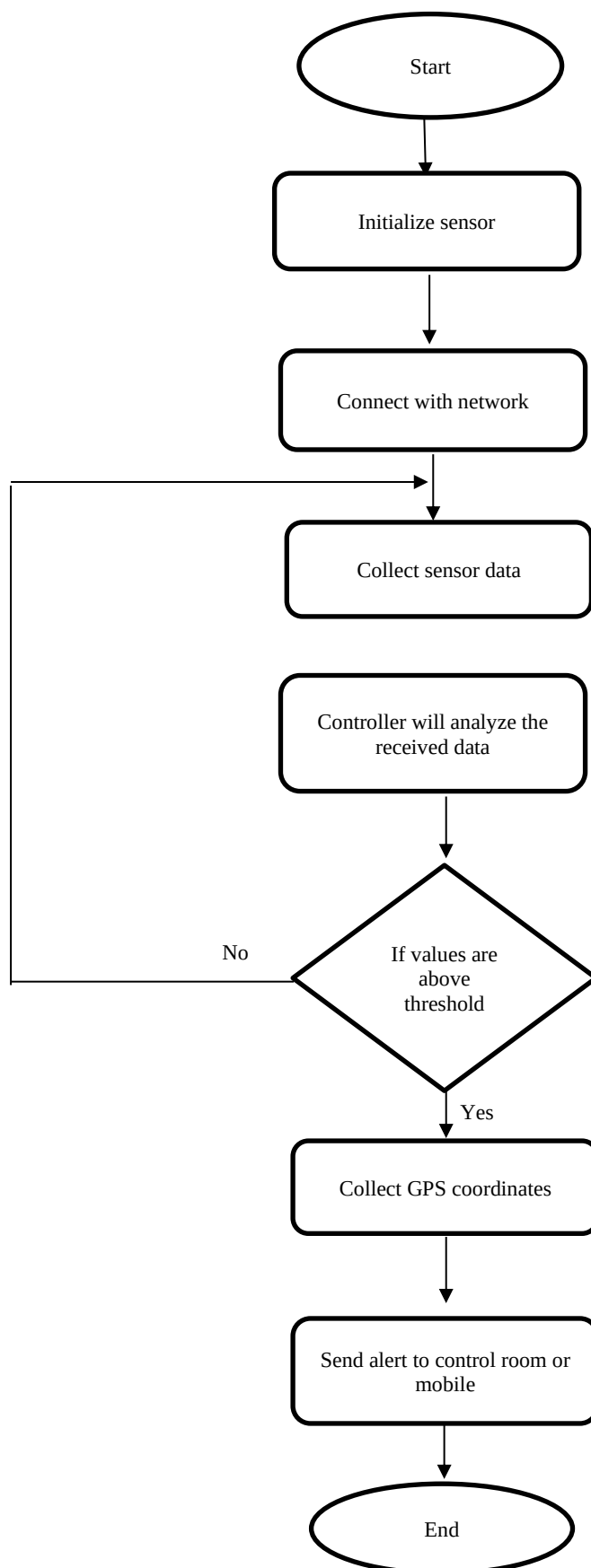


Figure 1. Flow chart of smart Helmet of Construction workers

RESULTS

Existing studies consistently report positive impacts on productivity attributed to IoT-based smart helmets. Features such as communication facilitation, access to real-time data, and task management, real-time hazard alerts and monitoring signs that contribute to accident prevention and early intervention in emergency situations. The review paper highlights common challenges such as connectivity issues. Recommendations to this include continuous improvement in hardware and software reliability to mitigate such issues. Suggestions for improved comfort features and customizable options are noted.

ABBREVIATIONS

IOT: Internet of Things,
GSM: Global system for mobile communication,
LoRA: Long range,
GPS: Global positioning system,
WAN: Wide area network,
LED: Light emitting diode,
SVM: Support vector machine

CONCLUSION

Internet of things based smart helmets for construction is regarded as a transformative approach for magnifying safety and productivity. With the use of integration of sensors and communication technologies these helmets offer real-time monitoring of workers all activities and environmental parameters. Features such as fall detection, work mode tracking, and GPS localization contribute to comprehensive safety protocols. Cloud connectivity enables smooth data exchange for timely decision-making. While user comfort remains a priority, further research could explore advanced sensor integration and predictive analytics. Standardization efforts would facilitate widespread adoption and integration with existing safety protocols. Overall, IoT based smart helmets hold immense potential to redefine safety standards and cultivate a culture of prevention in the construction industry, ultimately safeguarding the well-being of workers and improving operational efficiency.

Declaration of Interest

The authors of this review paper declare that they have no conflicts of interest related to this research project.

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